Advantage in Actor-Critic (A2C)

- The advantage function calculates the relative advantage of an action compared to others taken in that state; how taken an action in a state compares to the average val. in that state.

Advantage func.

$$A(s,a) = \underbrace{Q(s,a)}_{\text{q-val for that action}} - V(s) \text{ — average value of the state.}$$

$A(s,a) > 0$ - pushed in that direction

$A(s,a) < 0$ - pushed away from that direction.

$$A(s,a) = Q(s,a) - V(s)$$
$$= \underbrace{r + \gamma V(s') - V(s)}_{\text{TD error}}$$

Unit 7 MARL - Multi Agents Reinforcement Learning

Cooperative env - agents interact to maximize common benefits

Competitive / Adverserial env. - maximize its benefits by minimizing the opponent's

Mixed env. - agents have teams - some work together, others compete

Decentralized architecture - no information is shared b/w agents, they learn on their own

↳ non-stationary env. problem.

Centralized architecture - info is shared in a common buffer, learn from collective experience.

Agent cannot learn unless the opponent is at a similar level, self-play

Self-play is when the agent trains against previous copies of its own policy

Start w/ a copy of the agent, as it learns, update the opponent w/ a more recent policy.

Tradeoff b/w skill level, generality of policy, & stability of learning. Training w/ a set of slowly changing or unchanging adversaries → more stable training but there is a risk of overfitting

Hyperparameters:

- How often opponents are changed
- The # of opponents saved, ↑ = ↑ diversity
- The prob. of playing against self vs. less recent op
- The # of training steps before saving a new op.

ELO rating system - relative skill level between 2 players in a 0 sum game

↳ inferred from the losses & draws against other players, → player ratings depend on the ratings of their ops. & the results scored against them

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}$$

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}$$

at the end of the game, a linear adjustment is used to update the ELO

$$R'_A = R_A + K(S_A - E_A)$$

K is a constant (max adj. rating per game)

Advantages of ELO sys.

- pts. always balanced (sum the same)
- self-corrected sys. weaker player wins, ↑ pts. gained
- works w/ team games average for team used as elo.

Disadvantages

- doesn't take into account individual contributions of team members.
- Rating deflation - good rating requires skill over time to maintain.
- can't compare rating history